INTERNATIONAL A-LEVEL CHEMISTRY

(9620) CH04

Report on the examination

January 2024

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH04 JANUARY 2024

In general, the questions involving calculations were answered very well by students with many getting full marks for these questions.
Mechanisms were generally drawn very well although some students drew the curly arrow from an atom rather than the bond. Students need to be aware that curly arrows should be drawn from a lone pair of electrons or from a covalent bond.
The questions about practical techniques proved more difficult for many students. Students need to be more familiar with the required practicals so that they are able to describe why different steps are taken in these experiments and interpret results.

QUESTION 01

01.1 Many got full marks for this question. Those who lost marks did not show that $p(H_2) = p(I_2)$ or didn't square the partial pressure of HI in their calculations even though they had written this in their expression for K_p

01.2 Most got this correct. Some gave the units for K_c rather than K_p in their explanations.

01.3 Most students got the first mark for identifying that the forward reaction was endothermic (or reverse reaction was exothermic). About half of the responses did not explain why equilibrium would shift to the left and failed to mention opposing the temperature change so did not get the second mark for this question. Some students answered about the effect on rate or energy of particles rather than the position of equilibrium.

QUESTION 02

02.1 Most students got this correct. Common errors were giving the molecular formular rather than the empirical formula or giving the structural formula.

02.2 Most got both marks for this question. For those who got one mark, the mark lost was almost equally for classification of amine and for number of peaks. The most common error for classification was referring to an amide rather than amine.

02.3 Around 30% got full marks. The main error was on naming the carbonyl containing monomer. Around 50% students wrote pentan(-1,5-)dioic acid rather than pentane(-1,5-)dioic acid. Students should know that if there is a consonant following the alkane/alkene then the 'e' of the alkane/alkene is included in the name. Most students who gave the numbers of the dioc acid group/diacyl dichloride group gave them in the form -1.5- rather than -1,5- this was not penalised as the numbers were not required but students should use commas when giving more than one number in the names of compounds. Quite a few named the acid as pentane(-1,5-)dicarboxylic acid which was not credited.

02.4 Around 80% students got this question correct. The most common incorrect response was the polar C-N bonds in the chains.

02.5 Very few students got full marks on this question. Around 10% of students did not attempt to answer this question. A surprising number did not mention that bonds in nylon are polar. A number of students discussed hydrogen bonding between the water molecules and the polymer chains breaking the intermolecular forces. Many discussed the effect of high temperature on the rate of reaction rather than discussing why the polymer would be hydrolysed.

QUESTION 03

03.1 Most students got both marks in this question. Those that only got one mark usually did not draw the skeletal formula of the product of the reaction correctly – some drew displayed formulas and others drew the propanal or propanone rather than propanoic acid. Most were able to recall the colour change although a few only gave the final colour rather than the colour change.

03.2 Around 90% students answered this correctly.

03.3 Around 50% students achieved both marks for this question. The most common error was in naming the product – some had 'hydroxyl' rather than 'hydroxy'; some numbered the hydroxy group incorrectly; and some did not include the 'e' at the end of butane in the name. Students should give the structure of an aldehyde as R-CHO (some wrote CH_3CH_2COH, this was not penalised in this context but would have been if the structure had been required).

03.4 Around 50% students answered this correctly. The vast majority of incorrect responses stated that a racemic mixture had been formed.

03.5 Over 90% students answered this correctly. The wavenumber range (or a value within the range) needed to be stated as the question had asked students to use Table A on the Chemistry Data Sheet.

03.6 Most students answered this correctly. The most common errors were stating that the iodine was in excess or simply stating iodine was zero order without explaining why.

03.7 Most students were awarded M1 for stating that H^+ is a catalyst. Not all students explained that this could be deduced as H^+ is in the rate equation but not in the overall equation. Students needed to mention that it is in the rate equation as well as commenting on it not being in the overall equation.

QUESTION 04

04.1 Fewer than 50% students were awarded both marks for this question. The most common error was not including the curly arrow from the C=O to oxygen in the first step. Some students did not draw the curly arrows from the O–H bond to O^+ or the C–Cl bond to Cl in the intermediate instead drawing arrows from the H or Cl atoms. Others did not draw the curly arrow from a lone pair of electrons on O^- to form C=O

04.2 Most students answered this correctly. The most common error was not calculating the M_r of aspirin correctly.

04.3 More students did not attempt this question than any other on the paper. It was clear that many students had learnt the steps involved in recrystallisation, but some were not able to apply the information given in the question about the relative solubility of aspirin in ethanol and water. This meant that some students answered that the aspirin would be dissolved in a minimum volume of hot water in the first step and/or that the aspirin is washed in ethanol in the last step.

04.4 Around 30% got this question correct. Many gave the product of the reaction as ethanal rather than ethanoic acid. A few students drew the structure of ethanoyl chloride as a reactant rather than ethanoic anhydride.

QUESTION 05

05.1 Most students got this correct. The most common error was stating that it was to speed up the reaction.

05.2 Most students answered this correctly. Some students' answers implied that they thought that this was a stage in a recrystallisation process.

05.3 Almost all students gave an incorrect answer to this question. Many thought that the NaOH was added to quench the reaction. Some thought it was to do with the position of equilibrium.

05.4 Most students were awarded three marks for this question. Marks were most commonly lost for scales that only occupied part of the graph paper and for drawing straight lines between the plotted points rather than a line of best fit.

05.5 A surprising number of students did not draw a tangent to the curve at 30 minutes and instead divided the concentration at 30 mins by the time so only got one mark for the units. Some of those who drew a tangent were not able to use their scale on the graph to identify correct values to calculate the gradient.

05.6 Almost all students were awarded M1 for deducing that the reaction was first order. A large number stated that the line was linear but did not add that it went through the origin so were not able to get M2.

QUESTION 06

06.1 Around half of students got all four marks for this question. A few students drew the mechanism for making the electrophile rather than the mechanism for the reaction of the electrophile with benzene. The most common errors with the mechanism were not drawing the first curly arrow from the delocalised ring structure; drawing the horseshoe structure centred on the wrong carbon; and drawing the final curly arrow from the H atom rather than the C-H bond.

06.2 Simply stating that benzene has a delocalised ring structure that makes it stable was not enough to be awarded this mark. Students needed to state that this would be broken in an addition reaction or that it is preserved in a substitution reaction. A few students thought that the ring structure would repel an electrophile so were clearly confused about the definition of an electrophile.

06.3 Most students got this correct. Students need to remember to include that the substitution happens by a (free) radical mechanism.

06.4 Just over a third of students thought that the ammonia reacted only as a nucleophile. Almost all of the rest of the students answered correctly.

06.5 The majority of students gave the correct structure for C. There are still some students who forget to write that the nitric and sulfuric acids need to be concentrated. A few did not have the correct formula for nitric acid, and some gave reagents for Reaction 4 rather than Reaction 3.

06.6 Most students answered this correctly. Some thought that the M_r would be similar rather than the same as they had incorrectly worked out the number of hydrogens for one of the compounds. Stating that B and D were isomers of each other was not enough to be awarded the mark as students needed to note that mass spectrometry identifies the mass (of molecular ions and fragments).

QUESTION 07

07.1 Most students were able to write the equation correctly. The most common error was writing the reaction with sodium hydroxide rather than sodium hydrogencarbonate. A few students incorrectly showed the organic salt formed with a covalent bond between the oxygen of the carboxylate group and the sodium. The ionic attraction did not need to be shown but covalent bond was penalised.

07.2 Most students named the amino acid correctly. Those that did not either had the wrong numbers or thought it was 2-amino-3-methylbutanoic acid (i.e. they did not count the C of the carboxylic acid group in the carbon chain).

07.3 Very few students got both marks for this question. Many were able to draw the structure of the zwitterion but did not refer to the structure being ionic in their explanation of solubility.

07.4 Around 75% students got this correct. Students need to measure to the centre of the spot; some were measuring to the top or bottom of the spot.

07.5 Around 30% students answered this correctly. Some seemed to be unaware that a spot in chromatography can be a mixture with components having similar R_f values.

07.6 Most students were able to explain this. A few students answered incorrectly that there were more spots in Solvent 2 than in Solvent 1.

07.7 Most students answered this correctly. Those that got it wrong tended to discuss the bonding / interactions within the structure rather than classifying the structure.

07.8 Only around 20% got both marks for this question. Students needed to state that the cisplatin bonded to the N atom of the DNA (rather than simply that it bonded to the DNA) and also that the replication of DNA was inhibited. Some stated that the cell replication was inhibited but this information was given in the question so was not creditworthy.

07.9 Most answered this correctly.

QUESTION 08

08.1 Nearly 80% answered this question correctly and students had clearly learnt the structure of TMS well.

08.2 Surprisingly only just over 50% got both marks for this question despite it being a relatively common AO1 question on CH04. Some students restated that TMS was non-toxic (given in the question), and others referred to it being ‘stable’ rather than inert / unreactive.

08.3 Most students answered this correctly.

08.4 Most students answered this correctly. The most common error was drawing a benzene ring rather than cyclohexane.

08.5 Fewer than half of students answered this correctly. Students needed to mention that the dipoles in the bonds cancelled out, many did not mention dipoles.

08.6 Around 60% students got this correct. Some students do not know the correct terminology for splitting patterns so referred to quadret rather than quartet.

08.7 Most students answered this correctly and were able to give both structures. Around 85% gave at least one correct structure.

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